Renal Physiology

Renal Clearance

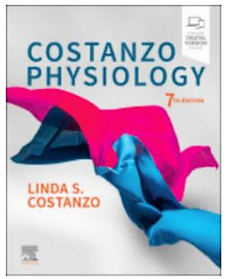
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Fall 2026

Learning Objectives

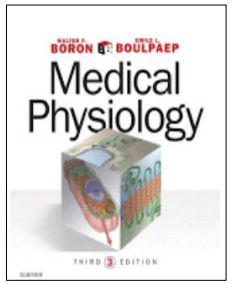
1. Define renal plasma clearance (Cx) and calculate clearance rates and filtered loads (FL) using standardized clinical laboratory values.
2. Explain how inulin clearance quantifies glomerular filtration rate (GFR) and evaluate why serum creatinine clearance serves as its clinical estimate.
3. Apply para-aminohippuric acid (PAH) clearance to calculate renal plasma flow (RPF) and determine total renal blood flow (RBF) using hematocrit.
4. Compare the clearance of freely filtered substances relative to GFR to determine whether the nephron exhibits net tubular reabsorption or net tubular secretion.
5. Analyze the carrier-mediated renal handling of glucose, defining transport maximum (Tm), renal threshold, and splay during progressive hyperglycemia.

Resources

READ



Chapter 6



Chapters 33 & 34

DO

* **TopHat Questions**
- **Guyton and Hall Review Questions (Library)**
- **Costanzo Cases**

TOP HAT

Join Code: 302237

Concept & Definition of Renal Clearance

- **Clearance Defined:** The *volume of plasma* that is completely cleared of a solute by the kidneys per unit time (mL/min or L/day).

- **Conceptual Principle:** Clearance does not measure the mass of solute removed; it measures the **volume of blood plasma** that contained that solute.

- **The Master Renal Clearance Equation:**

- $\boldsymbol{C_x = \frac{U_x \times \dot{V}}{P_x}}$

  - $U_x \times \dot{V}$ = Urinary Excretion Rate (mass/time)
  - P_x = Plasma Concentration (mass/volume)

- **Boundary Limits:** Renal clearance can never be negative, and for freely filtered solutes, it can never exceed total renal plasma flow (~ 625 mL/min).

Filtered Load vs. Excretion Rate

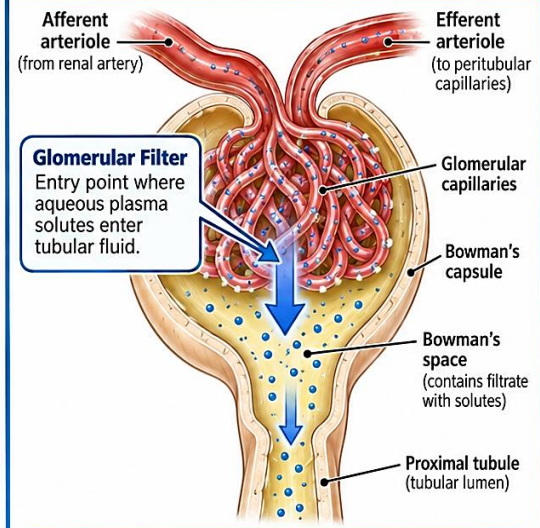
- **Filtered Load (FL) Defined:** The total quantity of a substance filtered into Bowman's space per unit time.

- **Filtered Load Equation (for freely filtered solutes):**
  - $FL_x = \text{GFR} \times P_x$

- **Excretion Rate Equation:**
  - $E_x = U_x \times \dot{V}$

- **Net Tubular Handling Deduced from Mass Balance:**
  - If $FL_x > E_x \rightarrow$ **Net Tubular Reabsorption** has occurred ($R = FL - E$).
  - If $FL_x < E_x \rightarrow$ **Net Tubular Secretion** has occurred ($S = E - FL$).
  - If $FL_x = E_x \rightarrow$ **No Net Tubular Transport** (filtration only).

Filtered Load



Filtered Load (FL) = GFR × P_x

Mass of solute filtered per unit time

mg/min or mEq/day

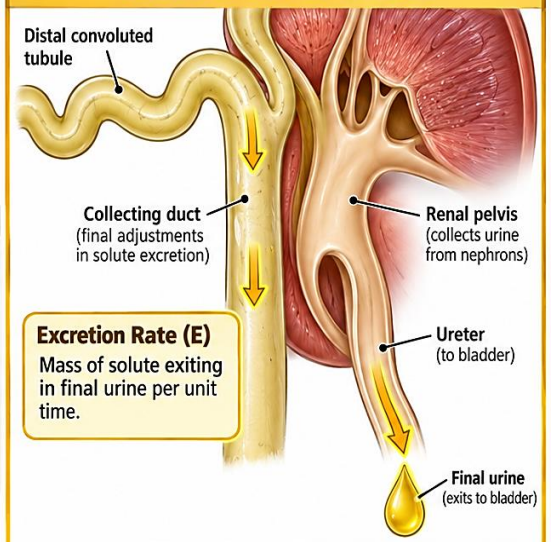
Filtration adds solute (x) to the tubular fluid at the glomerulus.

Comparison

Filtered Load = input to tubular fluid

Excretion Rate = output in final urine

Excretion Rate



$E = U_x \times \dot{V}$

Mass of solute excreted per unit time

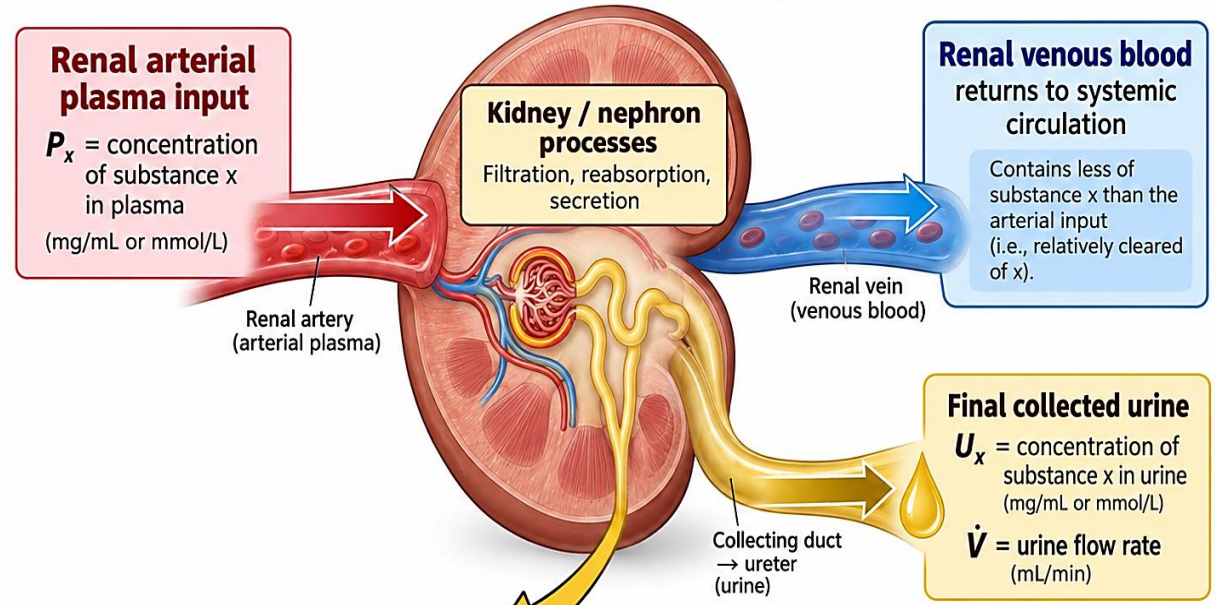
mg/min or mEq/day

Excretion is the final urinary output of solute (x) in the urine.

Reabsorption and secretion determine how final excretion compares with the filtered load.

Renal Clearance of a Substance (x)

Vascular → Tubular Input/Output Diagram



Net urinary elimination per minute (excretion rate)

Excretion rate of x = $U_x \times \dot{V}$

(mg/min or mmol/min)

Renal clearance of x

$$C_x = \frac{U_x \times \dot{V}}{P_x}$$

(mL/min)

Key point:

Clearance (C_x) is a virtual plasma volume per unit time, whereas $U_x \times \dot{V}$ is the excretion rate (amount of substance x eliminated per unit time).

A 50-year-old male with a normal GFR of 120 mL/min participates in a pharmacokinetic trial. A timed urine collection over 2 hours yields 180 mL of urine. Blood and urine sample results are the following: Plasma concentration of drug $X(P_x)$ = 0.5 mg/mL; Urine concentration of drug $X(U_x)$ = 20.0 mg/mL. *Determine Urine Flow Rate, Renal Clearance, Filtered Load, and Net Transport Direction.*

- **Step 1: Determine Urine Flow Rate ($\dot{V}$):**
  - $\dot{V} = \frac{180\ \text{mL}}{120\ \text{min}} = 1.5\ \text{mL/min}$
- **Step 2: Calculate Renal Clearance (C_x):**
  - $C_x = \frac{U_x \times \dot{V}}{P_x} = \frac{20.0\ \text{mg/mL} \times 1.5\ \text{mL/min}}{0.5\ \text{mg/mL}} = \frac{30.0}{0.5} = 60\ \text{mL/min}$
- **Step 3: Calculate Filtered Load (FL_x):**
  - $FL_x = \text{GFR} \times P_x = 120\ \text{mL/min} \times 0.5\ \text{mg/mL} = 60\ \text{mg/min}$
- **Step 4: Determine Net Transport Direction:**
  - $E_x = U_x \times \dot{V} = 20\ \text{mg/mL} \times 1.5\ \text{mL/min} = 30\ \text{mg/min}$
- Because $E_x(30\ \text{mg/min}) < FL_x(60\ \text{mg/min}) \rightarrow$ Net Tubular Reabsorption of 30 mg/min

Interim Summary 1: Clearance & Filtered Load

- **Clearance is a Virtual Volume:** Renal plasma clearance (C_x) does not quantify the physical mass of solute removed; it measures the **volume of arterial plasma** completely cleared of substance x per unit time (mL/min).

- **The Master Clearance Equation:** $C_x = \frac{U_x \times \dot{V}}{P_x}$, where the numerator ($U_x \times \dot{V}$) is the urinary excretion rate (mass/time) and the denominator (P_x) is plasma concentration (mass/volume).

- **Filtered Load (FL_x):** The total mass of a freely filtered solute entering Bowman's space per minute: $FL_x = \text{GFR} \times P_x$

- **Deducing Net Tubular Transport:**
  - $FL_x > E_x \rightarrow$ **Net Tubular Reabsorption** ($R = FL - E$)
  - $FL_x < E_x \rightarrow$ **Net Tubular Secretion** ($S = E - FL$)
  - $FL_x = E_x \rightarrow$ **No Net Transport** (filtration only)

Learning Objectives

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Inulin Clearance: The Gold Standard for GFR

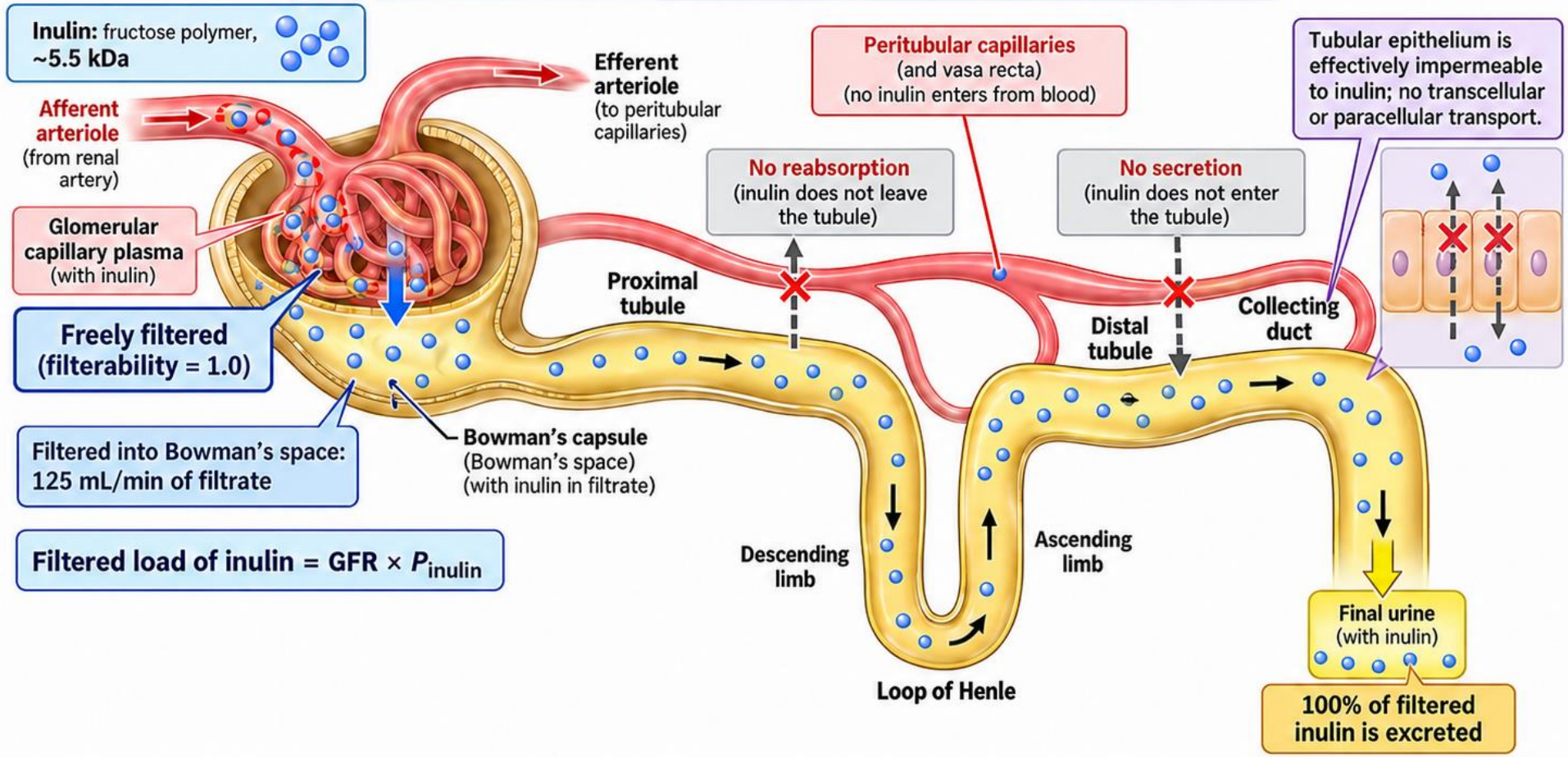
- **Properties of an Ideal GFR Marker:**
  - Freely filtered at the glomerulus (not restricted by size or charge).
  - Neither reabsorbed nor secreted by the renal tubules.
  - Biologically inert, non-toxic, and not metabolized or synthesized by the kidney.

Mathematical Proof that C_{inulin} = GFR:

- Filtered Load = Excretion Rate
- GFR$\times P_{inulin} = U_{inulin} \times \dot{V}$
- GFR $= \frac{U_{inulin}}{P_{inulin}} \times \dot{V} = C_{inulin}$

- **Clinical Limitation:** Inulin is an exogenous plant polysaccharide; measuring its clearance requires continuous intravenous infusion and catheterization, making it impractical for routine clinical care.

Inulin Transit Through the Nephron



KEY POINTS

1. Inulin is freely filtered at the glomerulus (filterability = 1.0).
2. Inulin is neither reabsorbed nor secreted anywhere along the nephron.
3. 100% of filtered inulin is excreted in the urine.
4. Therefore, clearance of inulin = GFR.

AMOUNTS TRANSPORTED

Filtered load of inulin = GFR × P_{inulin}
(= amount filtered per unit time)

Reabsorbed = 0

Secreted = 0

Excretion rate of inulin = $U_{inulin} \times \dot{V}$
(= amount excreted per unit time)

KEY EQUIVALENCE

Filtered load of inulin = Excretion rate of inulin

Therefore, clearance of inulin = GFR
(= ~125 mL/min in a normal adult)

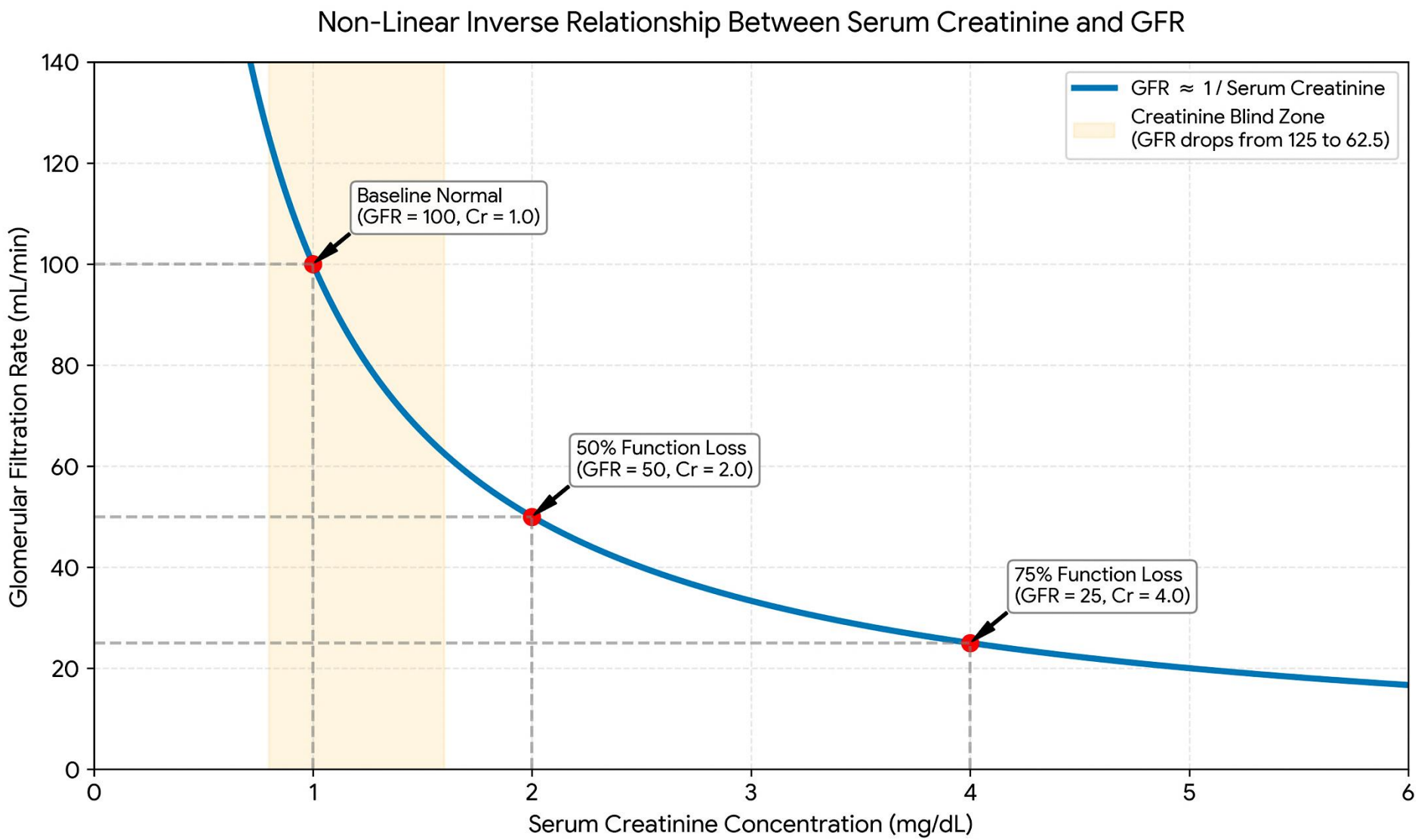
Creatinine Clearance: The Clinical Estimate of GFR

- **Origin of Creatinine:** Endogenous byproduct of muscle creatine breakdown released into circulation at a steady rate proportional to muscle mass.

- **Handling by the Nephron:**
  - Freely filtered across the glomerulus.
  - Not reabsorbed along the tubules.
  - **Slightly secreted** **(**$\sim 10{-}20\%$**)** by proximal tubule organic cation transporters.

- **Systematic Error in Clinical Estimation:**
  - Tubular secretion adds extra creatinine to the urine ($E > FL$).
  - Therefore, $C_{creatinine}$(~ 140 mL/min) **slightly overestimates true GFR** (~ 125 mL/min) by $10{-}20\%$.

- **Clinical Utility of Serum Creatinine (P_{Cr}):** At steady state, GFR is inversely proportional to serum creatinine ($P_{Cr} \approx \frac{1}{\text{GFR}}$). A doubling of serum creatinine represents a roughly 50% loss of functioning GFR.


Non-Linear Inverse Relationship Between Serum Creatinine and GFR
Glomerular Filtration Rate (mL/min)
Serum Creatinine Concentration (mg/dL)
GFR ≈ 1 / Serum Creatinine
Creatinine Blind Zone
(GFR drops from 125 to 62.5)
Baseline Normal
(GFR = 100, Cr = 1.0)
50% Function Loss
(GFR = 50, Cr = 2.0)
75% Function Loss
(GFR = 25, Cr = 4.0)
0
20
40
60
80
100
120
140
0
1
2
3
4
5
6

Interim Summary 2: Inulin, Creatinine, & GFR

- **Inulin is the Gold Standard for GFR:** Because inulin is freely filtered and undergoes **zero tubular reabsorption and zero tubular secretion**, its excretion rate equals its filtered load, making $C_{\text{inulin}} = \text{GFR} \approx 125$ mL/min.

- **Creatinine is the Clinical Surrogate:** Creatinine is an endogenous byproduct of muscle creatine breakdown that requires no IV infusion.

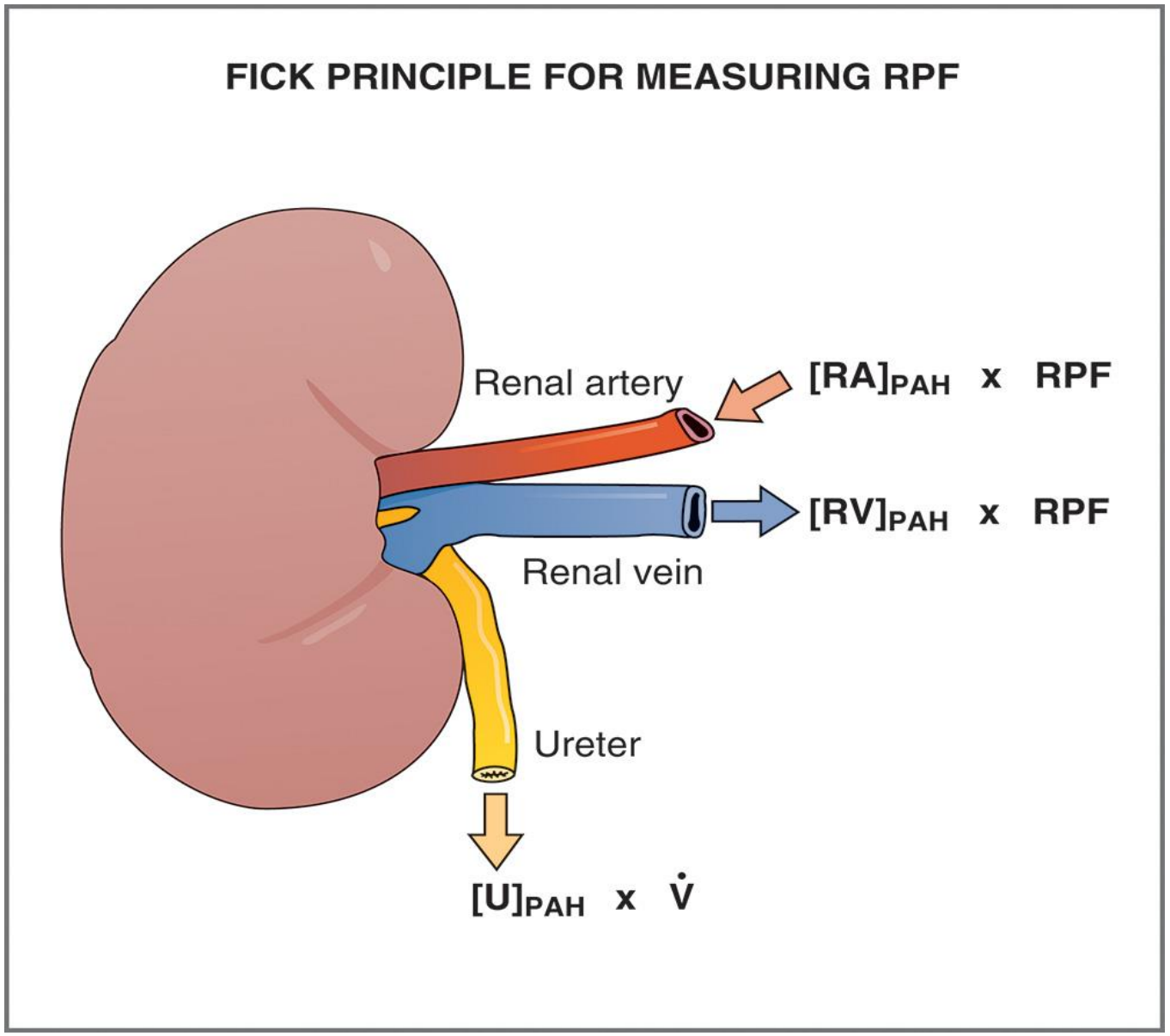
- **The Secretion Bias:** Because proximal tubular organic cation transporters slightly secrete creatinine ($\sim 10\text{–}20\%$), creatinine clearance **slightly overestimates true GFR** ($C_{\text{Cr}} \approx 140$ mL/min).

- **GFR and Serum Creatinine:** The relationship between serum creatinine (P_{Cr}) and GFR is inverse and non-linear ($P_{\text{Cr}} \approx \frac{1}{\text{GFR}}$). In early renal decline, GFR can drop by 50% (from 120 to 60 mL/min) while serum creatinine only rises subtly from 0.8 to 1.6 mg/dL.

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Measuring True Renal Blood Flow



RPF*([RA]pah - [RV]pah) =

$$RPF = \frac{[U]_{PAH} \times \dot{V}}{[RA]_{PAH} - [RV]_{PAH}}$$

RPF = Renal plasma flow

$[U]_{PAH}$ = [PAH]in urine

$\dot{V}$ = Urine flow rate

$[RA]_{PAH}$ = [PAH] in renal artery

$[RV]_{PAH}$ = [PAH] in renal vein

$$\textbf{True } \boldsymbol{RPF} = \frac{\boldsymbol{C_{PAH}}}{\textbf{PAH extraction ratio}}$$

Properties of Para-Aminohippuric Acid (PAH)

- **Properties of Para-Aminohippuric Acid (PAH):**
  - Freely filtered at the glomerulus.
  - Not reabsorbed by the tubules.
  - **Extensively secreted from peritubular capillaries** into the proximal tubule via organic anion transport (at low plasma concentrations).

- **Why PAH Measures Renal Plasma Flow:**
  - In a single pass through the kidney, virtually all PAH delivered to renal capillaries is transferred into the tubular lumen.
  - PAH is almost completely extracted from plasma passing through peritubular capillaries supplying proximal tubules capable of PAH secretion

- **To calculate eRPF:** *Effective RPF (eRPF) $\approx C_{PAH} = \frac{U_{PAH} \times \dot{V}}{P_{PAH}} \approx 625$ mL/min

***Roughly 10% of renal blood perfuses non-filtering capsular, medullary, and adipose tissue; thus, C_{PAH} underestimates absolute true RPF by about 10%.**

Converting RPF to Total Renal Blood Flow (RBF)

- **RBF Contains Both Plasma and Cells:** PAH clearance measures only the plasma fraction perfusing the kidneys (RPF).

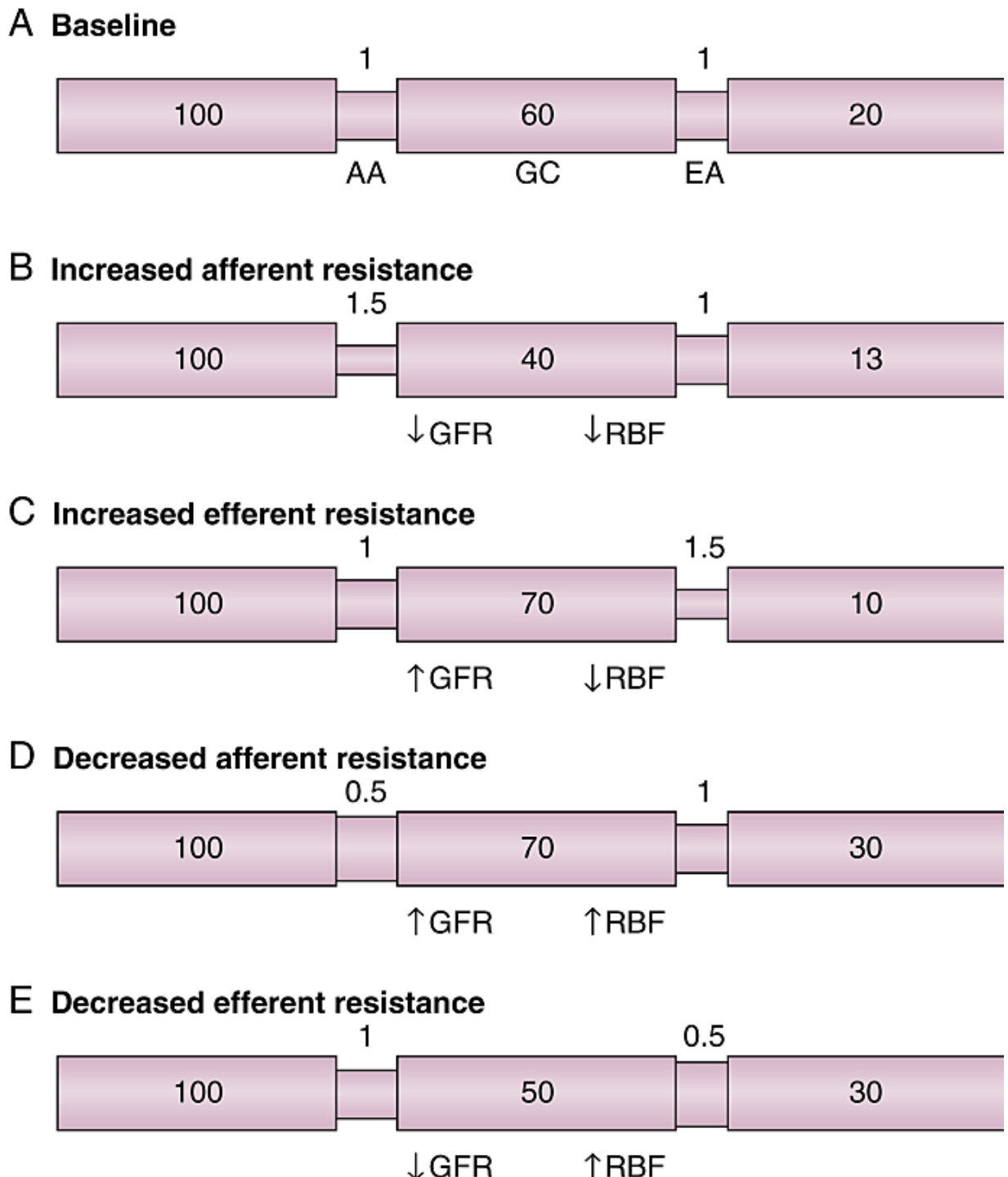
- **Calculating Total Renal Blood Flow:** $\text{RBF} = \frac{\text{RPF}}{1-\text{HCT}}$

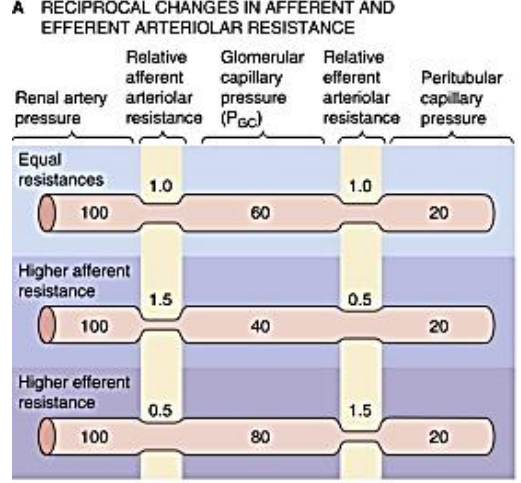
- **Filtration Fraction (FF):** The fraction of plasma flowing through the kidney that is filtered into Bowman's space.

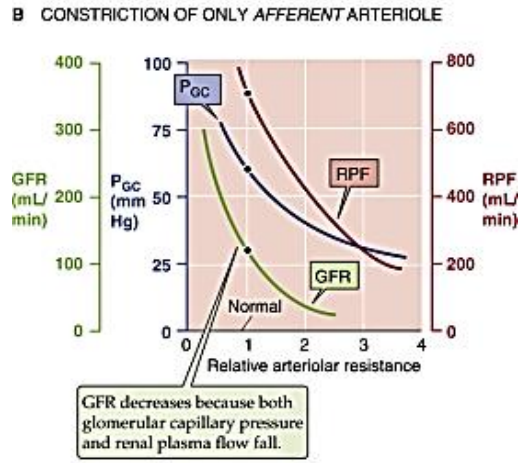
  - Formulated as: $\text{FF} = \frac{\text{GFR}}{\text{RPF}} = \frac{C_{inulin}}{C_{PAH}}$

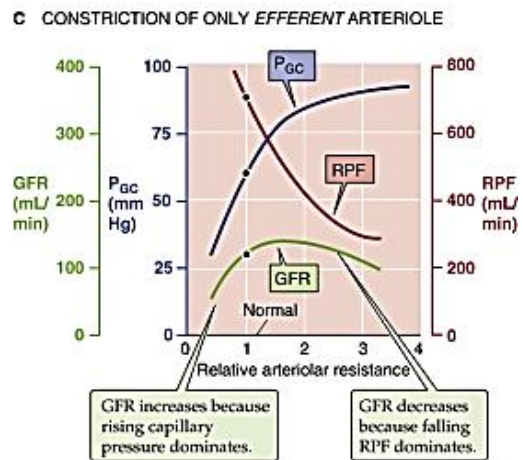
  - **Normal Baseline:** $\text{FF} = \frac{125\ \text{mL/min}}{625\ \text{mL/min}} = 0.20\ (20\%)$

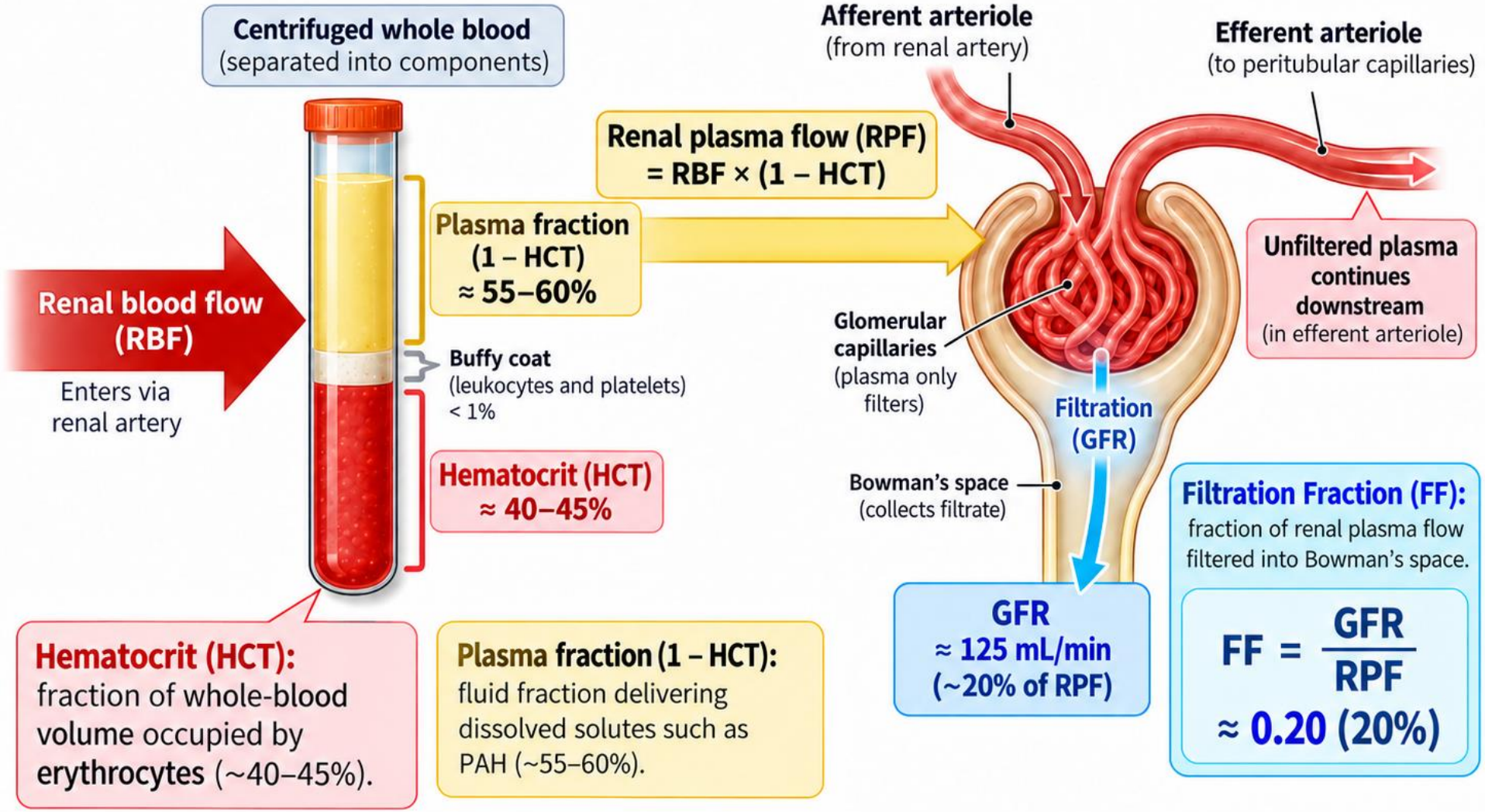
- Changes in arteriolar resistance alter FF:
  - Constricting efferent arteriole →↑ P_{GC} →↑ GFR, while ↓ RPF →↑↑ **FF**


A Baseline
1
1
100
60
20
AA
GC
EA
B Increased afferent resistance
1.5
1
100
40
13
↓GFR
↓RBF
C Increased efferent resistance
1
1.5
100
70
10
↑GFR
↓RBF
D Decreased afferent resistance
0.5
1
100
70
30
↑GFR
↑RBF
E Decreased efferent resistance
1
0.5
100
50
30
↓GFR
↑RBF


A RECIPROCAL CHANGES IN AFFERENT AND EFFERENT ARTERIOLAR RESISTANCE
Renal artery pressure
Relative afferent arteriolar resistance
Glomerular capillary pressure (P_{GC})
Relative efferent arteriolar resistance
Peritubular capillary pressure
Equal resistances
1.0
1.0
100
60
20
Higher afferent resistance
1.5
0.5
100
40
20
Higher efferent resistance
0.5
1.5
100
80
20


B CONSTRICTION OF ONLY AFFERENT ARTERIOLE
GFR (mL/min)
P_{GC} (mm Hg)
RPF (mL/min)
P_{GC}
RPF
GFR
Normal
Relative arteriolar resistance
GFR decreases because both glomerular capillary pressure and renal plasma flow fall.


C CONSTRICTION OF ONLY EFFERENT ARTERIOLE
GFR (mL/min)
P_{GC} (mm Hg)
RPF (mL/min)
P_{GC}
RPF
GFR
Normal
Relative arteriolar resistance
GFR increases because rising capillary pressure dominates.
GFR decreases because falling RPF dominates.



Only the plasma fraction contributes to RPF;
about 20% of renal plasma flow is filtered at the glomerulus.

A 52-year-old female is evaluated for suspected renovascular hypertension. During an intravenous infusion of PAH, steady-state parameters are obtained: GFR = 120 mL/min; Plasma [PAH]: (P_{PAH}) = 0.10 mg/mL; Urine [PAH]: (U_{PAH}) = 50.0 mg/mL; Urine flow rate: ($\dot{V}$) = 1.2 mL/min; Hematocrit: (HCT) = 40% (0.40). *Calculate renal plasma flow (RPF), renal blood flow (RBF), and filtration fraction.*

- **Step 1: Calculate Renal Plasma Flow (RPF via C_{PAH}):**
  - $\text{RPF} = C_{PAH} = \frac{U_{PAH} \times \dot{V}}{P_{PAH}} = \frac{50.0 \text{ mg/mL} \times 1.2 \text{ mL/min}}{0.10 \text{ mg/mL}} = \frac{60.0}{0.10} = 600 \text{ mL/min}$
- **Step 2: Calculate Total Renal Blood Flow (RBF):**
  - $\text{RBF} = \frac{\text{RPF}}{1-\text{HCT}} = \frac{600 \text{ mL/min}}{1-0.40} = \frac{600 \text{ mL/min}}{0.60} = 1000 \text{ mL/min (1.0 L/min)}$
- **Step 3: Calculate Filtration Fraction (GFR = 120 mL/min):**
  - $\text{FF} = \frac{\text{GFR}}{\text{RPF}} = \frac{120 \text{ mL/min}}{600 \text{ mL/min}} = 0.20 \ (20\%)$
- **Physiological Takeaway:** Her kidneys receive 1.0 L/min of blood (~ 20% of resting cardiac output). Of the 600 mL/min of plasma, 20% is filtered at the glomerulus, while 80% is routed to peritubular capillaries.

Interim Summary 3: PAH Clearance, RPF, & RBF

- **PAH Measures Effective RPF:** At low plasma concentrations, PAH is freely filtered (20%) and its remaining post-glomerular load is extensively secreted (80%) by proximal organic anion transporters in a single pass. Because renal venous blood is virtually cleared of PAH, $C_{\text{PAH}} \approx$ eRPF $\approx$ $600–625$ mL/min.

- **Converting Plasma Flow to Whole Blood Flow:** Red blood cells carry negligible PAH and are excluded from filtration. Total renal blood flow is calculated using hematocrit:
  - RBF $= \frac{\text{RPF}}{1-\text{HCT}} \approx \frac{600 \text{ mL/min}}{1-0.40} = 1000$ mL/min $(1.0$ L/min$)$

- **Filtration Fraction (**FF**):** The fraction of renal plasma flow that crosses into Bowman's capsule: $\text{FF} = \frac{\text{GFR}}{\text{RPF}} = \frac{C_{\text{inulin}}}{C_{\text{PAH}}} \approx 0.20\ (20\%)$

- **Arteriolar Tuning of FF:** Constricting the efferent arteriole increases glomerular capillary hydrostatic pressure (P_{GC}) while reducing total plasma flow, markedly driving up the filtration fraction (↑↑ FF).

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The Clearance Ratio (C_x/C_{inulin})

- **The Clearance Ratio Defined:** Fractional excretion of a substance relative to the filtration marker inulin:
  - **Clearance Ratio** $= \frac{C_x}{C_{inulin}}$
- **Three Interpretive Rules (for freely filtered solutes):**
  - $C_x/C_{inulin} < 1.0 \rightarrow$ **Net Reabsorption:** Solute was reclaimed from tubular fluid back into peritubular capillaries ($E_x < FL_x$).
  - $C_x/C_{inulin} = 1.0 \rightarrow$ **No Net Transport:** Solute was neither reabsorbed nor secreted; cleared strictly by ultrafiltration.
  - $C_x/C_{inulin} > 1.0 \rightarrow$ **Net Secretion:** Solute was actively added to the tubular lumen from peritubular blood ($E_x > FL_x$).
- **High-Yield Note:** "Net" transport means total reabsorption exceeds total secretion; it does not preclude bidirectional transport along different nephron segments (e.g., potassium, urea).

In a 24-hour period, 1.44 L of urine is collected from a man receiving an infusion of inulin. In his urine, the [inulin] is 150 mg/mL and the [Na^+] is 200 mEq/L. In his plasma, the [inulin] is 1 mg/mL and the [Na^+] is 140 mEq/L. *What is the clearance ratio for Na^+, and what is the significance of its value?*

- The clearance ratio for Na^+ is the clearance of Na^+ relative to the clearance of inulin. The clearance equation for any substance is $C = [U] \times \dot{V}/[P]$. All of the values needed are provided in the description, although urine flow rate ($\dot{V}$) must be calculated.

- $\dot{V}$ = Urine volume/time
  $= 1.44\text{L}/24\text{hr}$
  $= 1440\text{mL}/1440\text{min}$
  $= 1.0\text{mL/min}$

$$C_{Na^+} = \frac{[U]_{Na^+}}{[P]_{Na^+}}(\dot{V})$$
$$= \frac{200\text{mEq/L} \times 1\text{mL/min}}{140\text{mEq/L}}$$
$$= 1.43\text{mL/min}$$

$$C_{inulin} = \frac{[U]_{inulin}}{[P]_{inulin}}(\dot{V})$$
$$= \frac{150\text{mg/mL} \times 1\text{mL/min}}{1\text{mg/mL}}$$
$$= 150\text{mL/min}$$

- $$\frac{C_{Na^+}}{C_{inulin}} = \frac{1.43\text{mL/min}}{150\text{mL/min}}$$
  $= 0.01$ or 1%

- The clearance ratio of 0.01 means that only 1% of the filtered Na^+ is excreted. Stated differently, 99% of the filtered Na^+ must have been reabsorbed.

The Renal Clearance Spectrum

- **Sodium is Conserved:** With a clearance of less than 1 mL/min, the kidney reclaims $> 99\%$ of filtered sodium to protect blood pressure.

- **Secretion Accelerates Elimination:** Penicillin and PAH have clearances well above GFR, demonstrating that active peritubular secretion clears drugs far faster than filtration alone.

Substance	Clearance (C_x)	Clearance Ratio (C_x / C_{inulin})	Net Tubular Processing
Glucose	**0 mL/min**	**0**	**Freely filtered; completely reabsorbed**
Na^+	~0.9 mL/min	< 0.01	Freely filtered; >99% reabsorbed
Chloride	~1.3 mL/min	~0.01	Freely filtered; >99% reabsorbed
K^+	~12 mL/min	~0.10	Freely filtered; net reabsorbed (variable secretion in CD)
Inulin	**125 mL/min**	**1.0**	**Freely filtered; no reabsorption, no secretion (Anchor)**
Creatinine	**~140 mL/min**	**~1.12**	**Freely filtered; slightly secreted**
Penicillin	~350 mL/min	~2.8	Freely filtered; heavily secreted via proximal OAT
PAH	**625 mL/min**	**5.0**	**Freely filtered; extensively secreted (Measures RPF)**

Interim Summary 4: The Clearance Ratio

- **The Clearance Ratio as a Diagnostic Ruler:** Comparing solute clearance to inulin clearance ($\frac{C_x}{C_{\text{inulin}}}$) immediately establishes net nephron handling:
  - $\frac{C_x}{C_{\text{inulin}}} < 1.0 \rightarrow$ **Net Reabsorption** (e.g., Na^+, Cl^-, HCO_3^-, urea, glucose).
  - $\frac{C_x}{C_{\text{inulin}}} = 1.0 \rightarrow$ **Glomerular Marker Only** (Inulin).
  - $\frac{C_x}{C_{\text{inulin}}} > 1.0 \rightarrow$ **Net Secretion** (e.g., PAH, penicillin, K^+ under high-intake states).

- **Sodium Conservation:** Sodium clearance is less than 1 mL/min ($\frac{C_{Na}}{C_{\text{inulin}}} \approx 0.01$), reflecting $> 99\%$ tubular reabsorption to defend effective circulating volume.

- **Secretion Accelerates Elimination:** Peritubular active secretion clears xenobiotics and protein-bound drugs far faster than glomerular filtration alone ($C_{\text{penicillin}} \approx 350$ mL/min).

Learning Objectives

1. Define renal plasma clearance (Cx) and calculate clearance rates and filtered loads (FL) using standardized clinical laboratory values.
2. Explain how inulin clearance quantifies glomerular filtration rate (GFR) and evaluate why serum creatinine clearance serves as its clinical estimate.
3. Apply para-aminohippuric acid (PAH) clearance to calculate renal plasma flow (RPF) and determine total renal blood flow (RBF) using hematocrit.
4. Compare the clearance of freely filtered substances relative to GFR to determine whether the nephron exhibits net tubular reabsorption or net tubular secretion.
5. Analyze the carrier-mediated renal handling of glucose, defining transport maximum (Tm), renal threshold, and splay during progressive hyperglycemia.

Glucose Titration Curve: Carrier Saturation Kinetics

- **Normal State:** At fasting plasma glucose ($\sim 70{-}100$ mg/dL), all filtered glucose is actively reabsorbed by apical Na^+/glucose cotransporters (SGLT2/1) in the proximal tubule.
  - Clearance of glucose $= 0$ mL/min

- **Transport Maximum (T_m):**
  - The finite maximum transport rate of tubular carrier proteins when fully saturated.
  - For glucose in an adult, average $T_m = 375$ mg/min

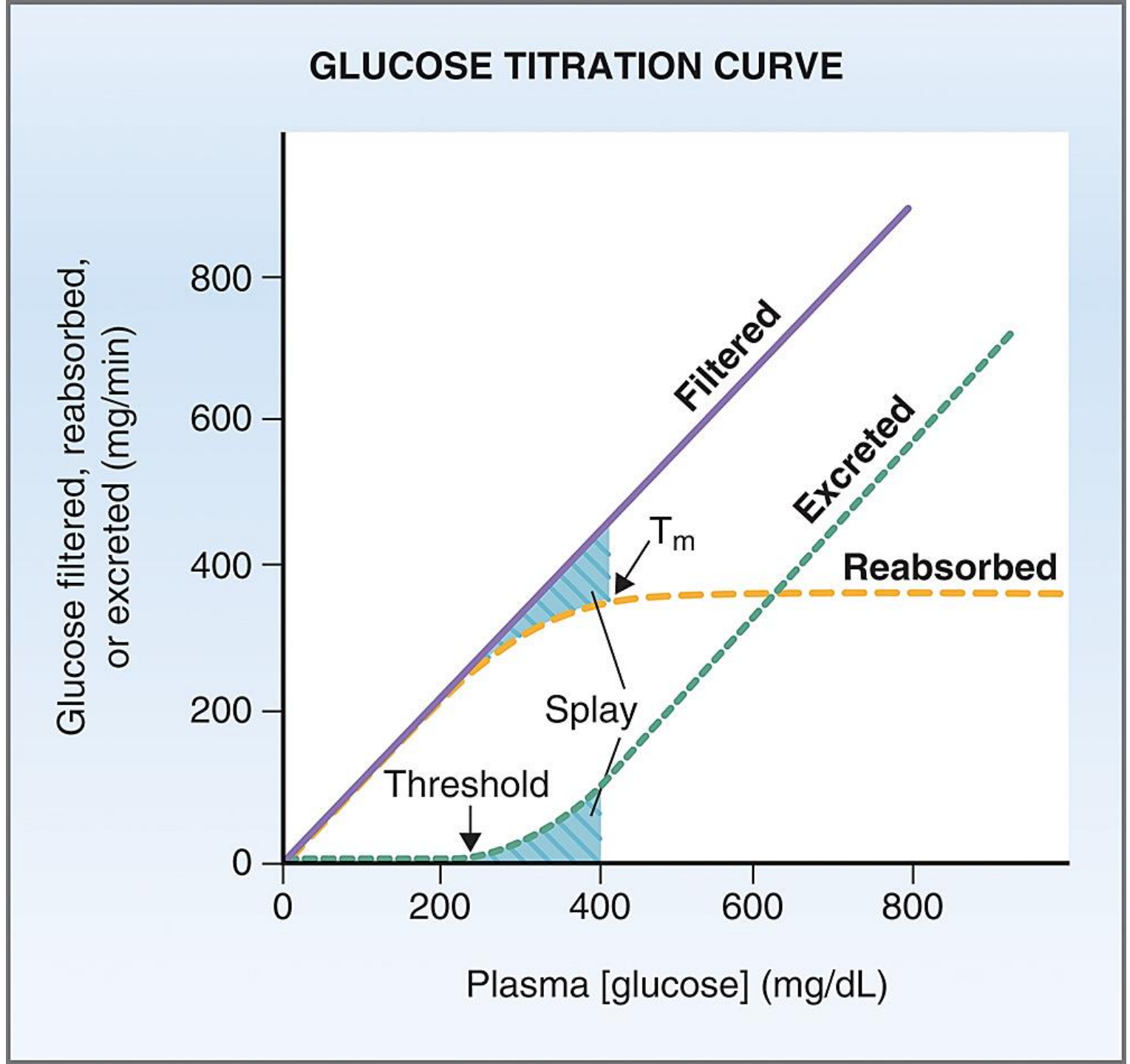
- **Governing Kinetic Rule:**
  - When Filtered Load $< T_m \rightarrow$ Reabsorption matches filtration; Excretion $= 0$.
  - When Filtered Load $> T_m \rightarrow$ Reabsorption is fixed at T_m; Excretion spills linearly into urine.

Renal Threshold, Splay, & Clinical Glucosuria

- **The Concept of Splay:** Glucose appears in the urine *before* the theoretical T_m is reached across the whole kidney.

- **Two Mechanisms Explaining Splay:**
  - *Nephron Heterogeneity:* Not all one million nephrons are anatomically identical; short proximal tubules with fewer SGLT carriers saturate at lower plasma levels than larger nephrons.
  - *Carrier Affinity / Kinetics:* As carriers near saturation, the binding affinity of SGLT decreases, permitting transient escape of glucose into tubular fluid down the nephron.

- **Pathophysiological Consequences (Diabetes Mellitus):**
  - Hyperglycemia exceeding threshold → Glucosuria.
  - Unreabsorbed luminal glucose acts as a non-reabsorbable osmotic solute, retaining water in the tubule.
  - Manifests clinically as **osmotic diuresis, polyuria, and compensatory polydipsia**.


GLUCOSE TITRATION CURVE
Glucose filtered, reabsorbed, or excreted (mg/min)
800
600
400
200
0
0
200
400
600
800
Plasma [glucose] (mg/dL)
Filtered
Excreted
Reabsorbed
T_m
Splay
Threshold

A 48-year-old male with poorly managed type 2 diabetes presents with fatigue and polyuria. Laboratory workup reveals: GFR = 125 mL/min; Plasma glucose ($P_{glucose}$) = 400 mg/dL (4.0 mg/mL); Proximal tubular glucose T_m: 375 mg/min. *Calculate the Filtered Load ($FL_{glucose}$) of glucose. How does this compare to Tm of glucose? Calculate glucose excretion rate ($E_{glucose}$) and the renal clearance of glucose ($C_{glucose}$).*

- **Step 1: Calculate Filtered Load of Glucose (FL_{glucose}):**
  - $FL_{\text{glucose}} = \text{GFR} \times P_{\text{glucose}} = 125 \text{ mL/min} \times 4.0 \text{ mg/mL} = 500 \text{ mg/min}$
- **Step 2: Compare Filtered Load to Transport Maximum (T_m):**
  - Because $FL_{\text{glucose}}(500 \text{ mg/min}) > T_m(375 \text{ mg/min})$, transport capacity is exceeded.
- **Step 3: Calculate Glucose Excretion Rate (E_{glucose}):**
  - $E_{\text{glucose}} = FL_{\text{glucose}} - T_m = 500 \text{ mg/min} - 375 \text{ mg/min} = 125 \text{ mg/min}$
- **Step 4: Calculate Renal Clearance of Glucose (C_{glucose}):**
  - $C_{\text{glucose}} = \frac{E_{\text{glucose}}}{P_{\text{glucose}}} = \frac{125 \text{ mg/min}}{4.0 \text{ mg/mL}} = 31.25 \text{ mL/min}$
- **Physiological Takeaway:** In non-diabetic states, glucose clearance is zero. Under severe hyperglycemia, clearance becomes positive (31.25 mL/min), reflecting significant urinary glucose wasting and resulting osmotic diuresis.

Interim Summary 5: Glucose Titration, T_m, & Splay

- **Carrier Saturation Kinetics:** Glucose reabsorption via apical SGLT transporters is carrier-mediated and saturable, reaching a maximum reabsorption capacity called the Transport Maximum ($T_m \approx 375$ mg/min).

- **Renal Threshold vs. T_m:** The theoretical saturation point occurs when filtered load reaches T_m (plasma glucose ~ 300 mg/dL), but glucose first appears in the urine at the **renal threshold** ($\sim 180{-}200$ mg/dL).

- **The Mechanism of Splay:** The rounded curve between threshold and T_m(splay) results from nephron heterogeneity (variations in nephron length and carrier density) and declining transporter binding affinity as carriers approach saturation.

- **Clinical Glucosuria:** When plasma glucose exceeds threshold, unreabsorbed luminal glucose acts as a non-reabsorbable osmotic solute, retaining tubular water and causing osmotic diuresis, polyuria, and compensatory polydipsia.

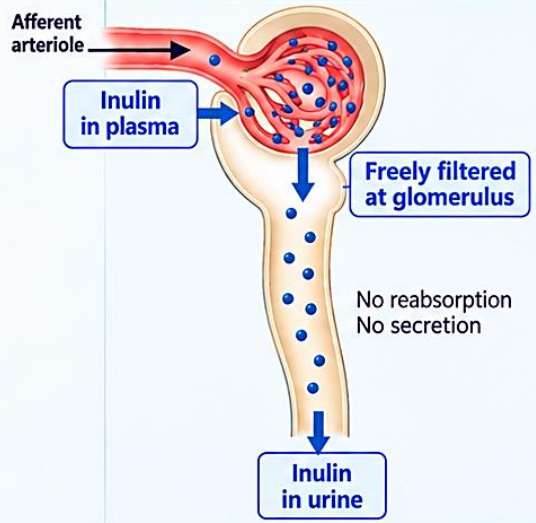
Renal Clearance Markers: Inulin vs PAH vs Glucose

Comparison of filtration, reabsorption/secretion, and what clearance represents.

INULIN

Filtered, neither reabsorbed nor secreted

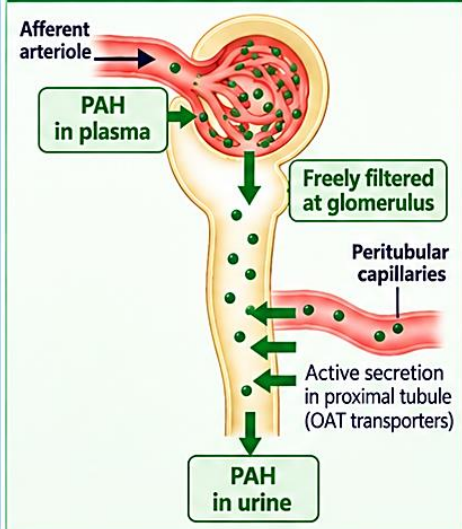
(C_{inulin} = GFR)



PAH

Filtered and actively secreted

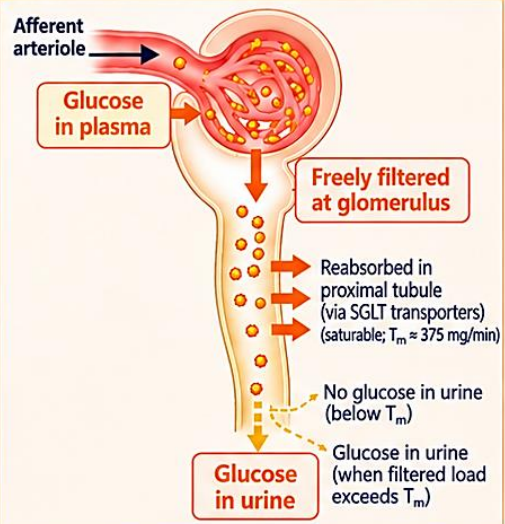
(C_{PAH} ≈ RPF)



GLUCOSE

Filtered and reabsorbed (saturable)

($C_{glucose}$ = 0 below T_m; > 0 when filtered load > T_m)



	Inulin	PAH	Glucose
Filtration	Freely filtered	Freely filtered	Freely filtered
Reabsorption	None	None (negligible)	Yes (saturable, via SGLT) T_m ≈ 375 mg/min
Secretion	None	Yes (extensive, proximal tubule via OAT transporters)	None
Clearance (C)	C_{inulin} = GFR	**C_{PAH} ≈ RPF** (almost completely removed in one pass)	$C_{glucose}$ = 0 (below T_m) > 0 when filtered load > T_m
What it measures	Glomerular filtration rate	Renal plasma flow (effective RPF)	Not used to measure GFR or RPF Illustrates reabsorptive capacity
Clinical/teaching notes	Exogenous marker (gold standard for GFR)		No glucose in urine until renal threshold (~180–200 mg/dL), then glucosuria when filtered load exceeds T_m

Master Summary

- **Clearance Master Formula:** $C_x = \frac{U_x \times \dot{V}}{P_x}$ measures the virtual volume of plasma cleared per minute.

- **Normal Clearance Values for Inulin/PAH:**
  - $C_{\text{inulin}} = \text{GFR} \approx 125$ mL/min (Clinically estimated by $C_{\text{creatinine}}$).
  - $C_{\text{PAH}} = \text{RPF} \approx 625$ mL/min $\rightarrow \text{RBF} = \frac{\text{RPF}}{1-\text{HCT}}$

- **Clearance Ratio (C_x/C_{inulin}):**
  - $< 1.0 \rightarrow$ Net Reabsorption
  - $> 1.0 \rightarrow$ Net Secretion

- **Glucose Transport:** Governed by SGLT saturation ($T_m = 375$ mg/min); threshold is lower (~ 200 mg/dL) due to splay.

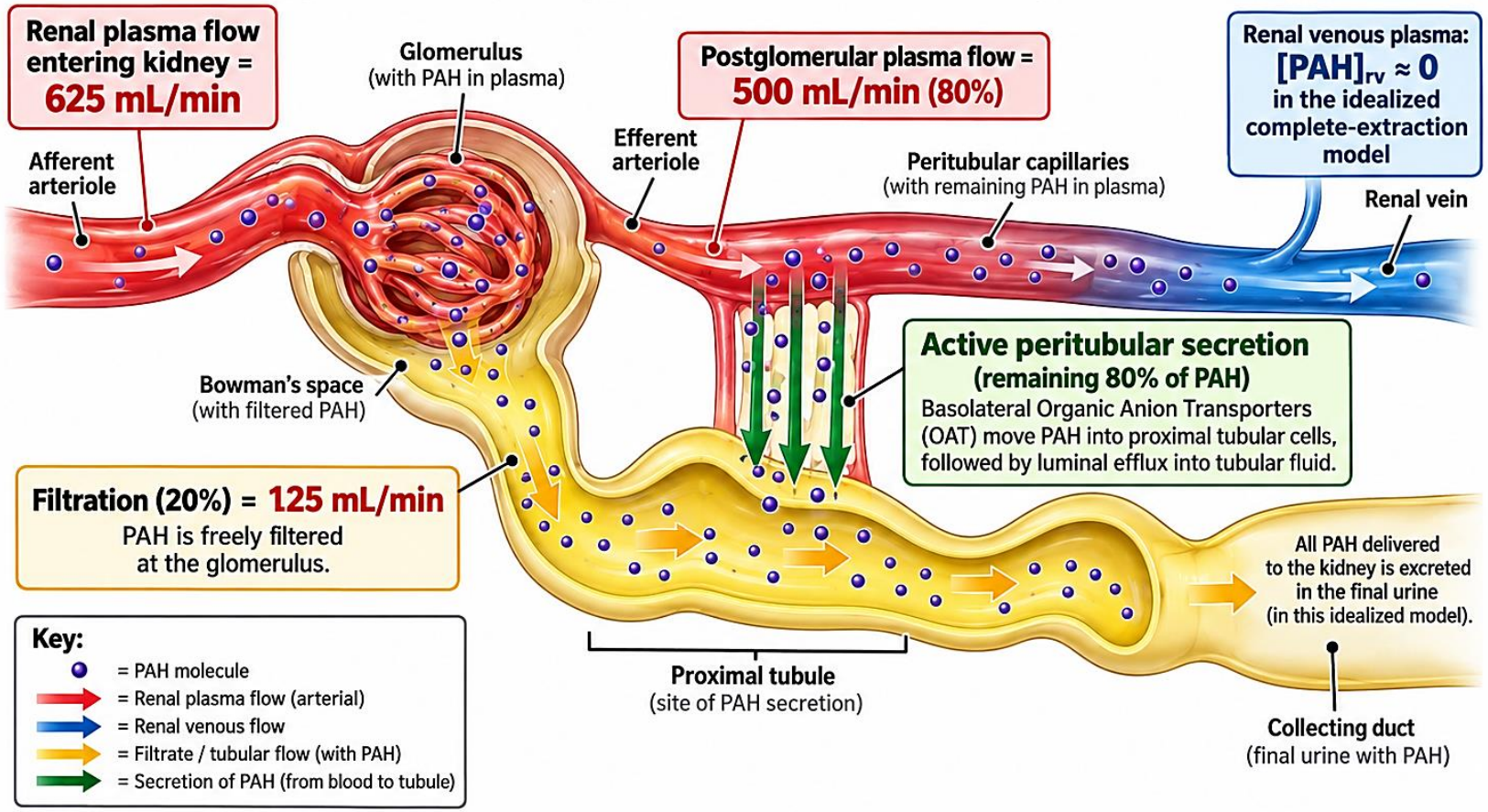
The End

Appendix Information

Reference materials, reading beyond the presentation, and anything else that needs to be included for completeness!

Idealized PAH Handling and Renal Plasma Flow

Why PAH clearance approximates effective renal plasma flow



In this idealized model:

Filtered PAH + Secreted PAH = Total PAH delivered in renal plasma

Therefore,

$C_{PAH} \approx$ effective RPF

Physiologic note:

In real kidneys, PAH extraction is high but not complete (renal venous [PAH] is low, not exactly zero), so PAH clearance estimates effective renal plasma flow (RPF) rather than exact true RPF.

Clearance Ratio

- C_x/C_{inulin} = 1.0. The clearance of x equals the clearance of inulin. The substance must also be a glomerular marker (filtered, but neither reabsorbed nor secreted).

- C_x/C_{inulin} < 1.0. The clearance of x is lower than the clearance of inulin. Either the substance is not filtered or it is filtered and subsequently reabsorbed.
  - For example, albumin is not filtered, and the clearance of albumin is less than the clearance of inulin. **The clearances of Na^+, Cl^-, HCO_3^-, phosphate, urea, glucose, and amino acids also are less than the clearance of inulin because these substances are filtered and then reabsorbed.**

- C_x/C_{inulin} > 1.0. The clearance of x is higher than the clearance of inulin. The substance is filtered and secreted. Examples of substances whose clearances are higher than that of inulin are organic acids and bases and, under some conditions, K^+.

Structure	Abbreviation	Meaning	Units and/or Normal Value
Whole Kidney	C	Clearance	mL/min
	[U]	Concentration in urine	mg/mL
	[P]	Concentration in plasma	mg/mL
	$\dot{V}$	Urine flow rate	mL/min
	GFR	Glomerular filtration rate	120 mL/min
	RPF	Renal plasma flow	660 mL/min
	RBF	Renal blood flow	1200 mL/min
Single Nephron	[TF]	Concentration in tubular fluid	mg/mL
	$[TF/P]_x$	Concentration of x in tubular fluid relative to concentration of x in plasma	None
	$[TF/P]_{inulin}$	Concentration of inulin in tubular fluid relative to concentration of inulin in plasma	None
	$[TF/P]_x/[TF/P]_{inulin}$	Fraction of the filtered load remaining in tubular fluid, or fractional excretion	None

TABLE 6.4

Commonly Used Equations in Renal Physiology

Name	Equation	Units	Comments
Clearance	$C_x = \frac{[U]_x V}{[P]_x}$	mL/min	x is any substance
Clearance ratio	$\text{Clearance ratio} = \frac{C_x}{C_{inulin}}$	None	Also means fractional excretion of x
Renal plasma flow	$RPF = \frac{[U]_{PAH} \dot{V}}{[RA]_{PAH} - [RV]_{PAH}}$	mL/min	
Effective renal plasma flow	$\text{Effective RPF} = \frac{[U]_{PAH} \dot{V}}{[P]_{PAH}}$	mL/min	Underestimates RPF by 10%; equals C_{PAH}
Renal blood flow	$RBF = \frac{RPF}{1 - Hct}$	mL/min	1 minus Hct is fraction of blood volume that is plasma
Glomerular filtration rate	$GFR = \frac{[U]_{inulin} \dot{V}}{[P]_{inulin}}$	mL/min	Equals C_{inulin}
Filtration fraction	$FF = \frac{GFR}{RPF}$	None	
Filtered load	Filtered load = GFR × $[P]_x$	mg/min	
Excretion rate	Excretion = $\dot{V} \times [U]_x$	mg/min	
Reabsorption or secretion rate	Reabsorption or secretion = Filtered load – Excretion	mg/min	If *positive,* net reabsorption If *negative,* net secretion
Free-water clearance	$C_{H_2O} = \dot{V} - C_{osm}$	mL/min	If *positive,* free water is excreted If *negative,* free water is reabsorbed

Hct, Hematocrit.